

# Signals and Systems — Solutions

CEN 211

## Question 1 — Derivative Property of LTI Systems

### Problem Statement:

An LTI system is characterized by the input–output relation  $x(t) \rightarrow y(t)$ . Find the output of the system when the input is  $x'(t)$ .

### Solution:

Let  $H$  denote the system operator, so that

$$H\{x(t)\} = y(t).$$

By definition, the derivative of the input is

$$x'(t) = \lim_{\Delta \rightarrow 0} \frac{x(t) - x(t - \Delta)}{\Delta}.$$

*Step 1 — Time-invariance.* A time shift of the input produces the same time shift of the output:

$$H\{x(t - \Delta)\} = y(t - \Delta).$$

*Step 2 — Linearity.* The system preserves scaling and addition, so applying  $H$  to the difference quotient gives

$$H\left\{\frac{x(t) - x(t - \Delta)}{\Delta}\right\} = \frac{H\{x(t)\} - H\{x(t - \Delta)\}}{\Delta} = \frac{y(t) - y(t - \Delta)}{\Delta}.$$

*Step 3 — Take the limit  $\Delta \rightarrow 0$ .* Passing to the limit on both sides,

$$H\{x'(t)\} = \lim_{\Delta \rightarrow 0} \frac{y(t) - y(t - \Delta)}{\Delta} = y'(t).$$

### Answer:

$$x'(t) \rightarrow y'(t) \quad \left(\text{i.e. } \frac{dy(t)}{dt}\right)$$

## Question 2 — Fundamental Period

### Problem Statement:

Find the fundamental period  $T_0$  of the continuous-time signal

$$y(t) = 7 \sin(400\pi t).$$

### Solution:

The standard form of a continuous-time sinusoidal signal is

$$y(t) = A \sin(\omega_0 t + \varphi).$$

By comparing the given signal to the standard form, the fundamental angular frequency is

$$\omega_0 = 400\pi \text{ rad/s}.$$

The fundamental period is related to the angular frequency by

$$T_0 = \frac{2\pi}{\omega_0} = \frac{2\pi}{400\pi} = \frac{1}{200} \text{ s}.$$

**Answer:**

$$T_0 = \frac{1}{200} \text{ s} = 5 \text{ ms}$$

*Note:* If the exam instead asks for the fundamental frequency,  $f_0 = \frac{1}{T_0} = 200 \text{ Hz}$ .

### Question 3 — Z-Transform

#### Problem Statement:

Find the Z-transform and the Region of Convergence (ROC) for the sequence

$$x[n] = a^n u[n].$$

#### Solution:

By definition, the bilateral Z-transform is

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}.$$

Since  $u[n] = 1$  for  $n \geq 0$ , the lower limit of the summation changes to  $n = 0$ :

$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n.$$

This is an infinite geometric series  $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$ , which converges if and only if  $|r| < 1$ . Setting  $r = az^{-1}$ , the convergence criterion (ROC) is

$$|az^{-1}| < 1 \implies \frac{|a|}{|z|} < 1 \implies |z| > |a|.$$

Applying the geometric-series sum formula within this region,

$$X(z) = \frac{1}{1 - az^{-1}} = \frac{z}{z - a}.$$

**Answer:**

$$X(z) = \frac{z}{z - a}, \quad \text{ROC: } |z| > |a|$$